BUILDING A USB RC METER

# Episode 5 Worksheet

## From Silicon to Screen: Firmware, UART, USB and the PC App

*Work through these after watching the episode. Show your working; answers are on the last page. Values come from the 2014 thesis data unless stated.*

### Learning objectives checked

- Read a peripheral register (ADCON0, T1CON, CM2CON0) bit by bit
- Compare polling with interrupts, and avoid the 'interrupt with no handler' trap
- Explain a UART frame, baud rate, and why TX goes to RX
- Decode the meter's text protocol, and choose a 2026 replacement for the Windows app

### Questions

1. Decode ADCON0 = 0xC5 on a PIC16F886 (bits: ADCS1 ADCS0 CHS3 CHS2 CHS1 CHS0 GO ADON).

2. Which register value selects channel AN4, FRC clock, ADC on, not converting?

3. At 9600 baud with 8-N-1 framing, how many characters per second can be sent? How long does "F47R19192" + CR take?

4. Draw (or describe) the UART waveform for the ASCII character 'R' (0x52).

5. Why is "asynchronous" not the same as "no clock needed"?

6. If the PIC ran at 4 MHz while the code assumed 8 MHz, what capacitance would the app report for a true 47 µF part?

7. What is the interrupt vector address on a PIC16 mid-range device, and what should be there?

8. Propose a checksum for the frame "2ndS233" and explain how the PC would use it.

## Answers

1. 0xC5 = 1100 0101: ADCS = 11 (FRC clock), CHS = 0001 (AN1), GO = 0, ADON = 1.
2. 1101 0001 = 0xD1 (the value used in the thesis firmware).
3. 9600 / 10 = 960 characters/s. 10 characters → ≈10.4 ms.
4. 0x52 = 0101 0010. Idle 1, start 0, then LSB first: 0 1 0 0 1 0 1 0, stop 1.
5. There's no shared clock WIRE, but both ends need their own accurate clocks set to the same baud rate to sample bits at the right times.
6. Each tick lasts 1 µs, but the app assumes 0.5 µs, so it computes half the real time: ≈23.5 µF (and the UART would run at ≈4800 baud).
7. 0x0004. It should hold the start of an interrupt service routine (or interrupts should stay disabled).
8. e.g. XOR all characters, then append '*' + two hex digits. The PC recomputes the XOR and discards the line if it doesn't match.

### Going further

**Microchip PIC16F882/3/4/6/7 datasheet (OSCCON, ADC, comparator, Timer1, EUSART):** http://ww1.microchip.com/downloads/en/devicedoc/41291f.pdf

**Microchip Developer Help: 8-bit PIC internal oscillator:** https://developerhelp.microchip.com/xwiki/bin/view/products/mcu-mpu/8bit-pic/oscillator-options/internal/

**Microchip PIC16F886 product page (status):** https://www.microchip.com/en-us/product/pic16f886

**Cytron UC00A (FTDI) and UC00C (CH340) USB–UART converters:** https://my.cytron.io/p-uc00c-ch340-usb-to-uart-converter

**pyserial documentation:** https://pyserial.readthedocs.io/

**MDN: Web Serial API:** https://developer.mozilla.org/en-US/docs/Web/API/Web_Serial_API

**Microsoft: Windows end of support:** https://www.microsoft.com/en-us/windows/end-of-support